SafeDrop
EWASTE

AWS ENVIRONMENTAL HACKS · WASTE AND ENERGY TRACK

Your old phone still holds your life.
Wipe it. Drop it. Done.

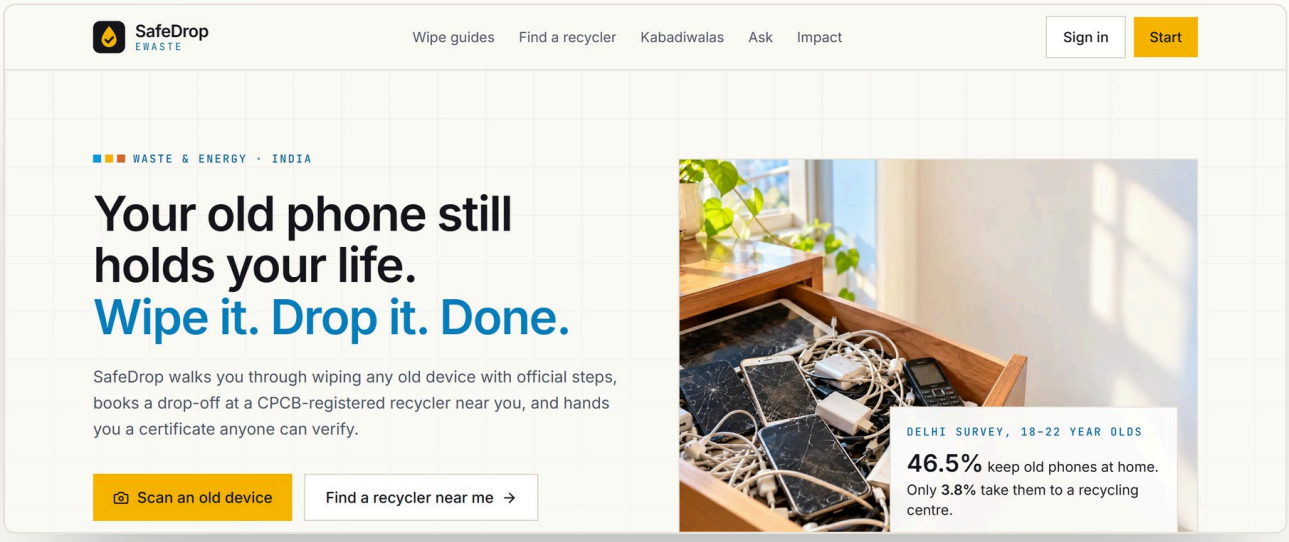
SafeDrop helps people in India clear out old phones, laptops and chargers safely. It gives an official wipe checklist for every device, finds a government-registered recycler or a verified kabadiwala pickup, and issues a certificate showing CO_2e avoided and energy saved.

573
CPCB-registered recyclers and refurbishers, geocoded

104
wipe steps in 10 guides, each linked to the maker's own page

15
features, all working live on AWS

0
made-up numbers: every figure has a source



Live app
main.d2ykddnngohyy3.amplifyapp.com

Demo login
demo@example.com
SafeDrop@2026

Source code
github.com/aditeejha/safedrop-ewaste

CONTENTS

What’s inside

01 The problem

02 What SafeDrop does

03 The 15 features

04 Scan and wipe

05 Find a recycler and book

06 Kabadiwala pickup partners

07 Handover and certificate

08 Ask SafeDrop: grounded AI

09 Waste to energy: how impact is counted

10 Built on AWS

11 Data model, privacy and safety

12 Real data and where it comes from

13 Testing and validation

14 Limits and next steps

15 Sources

Try it in two minutes. Open the live app and sign in with `demo@example.com` / `SafeDrop@2026`. One demo account is a citizen, a drop-point partner and a kabadiwala pickup partner, so one person can run the whole journey: add a device, wipe it, book, confirm the handover in the Partner portal and open the certificate. Demo activity is labelled and kept out of the public impact totals.

THE PROBLEM

01 Two fears keep devices in drawers

Old electronics pile up at home because people fear for their data and don’t know who will take a device safely. When devices do leave, many go to scrap dealers and into informal yards.

1,397,955 t

E-waste generated in India in FY 2024-25, as estimated by CPCB.

PIB · 11 Aug 2025

70.71%

Share reported as collected, dismantled and recycled or disposed formally that year. The rest is unaccounted for.

PIB · 24 Jul 2025

46.5%

Of 1,039 young adults surveyed in New Delhi keep old phones at home; only 3.8% take them to a recycling centre and 11.2% sell to scrap dealers.

Borthakur & Singh · 2022

26% · 19%

Of people keeping old phones worry about others accessing their data; 19% don’t know how to wipe it (UK survey; no Indian equivalent was found).

YouGov · May 2024

Why it matters

WHO says open burning and heating are the most hazardous informal e-waste activities. It links them to stillbirth, premature birth, neurodevelopmental effects from lead, and reduced lung function. Unsound recycling can release up to 1,000 chemical substances. The last official estimate, from a 2011 Rajya Sabha report, put more than 90% of India’s e-waste in the unorganised sector.

The law already points the right way. The E-Waste (Management) Rules, 2022 (in force since 1 April 2023) require e-waste to reach registered producers, refurbishers or recyclers. Loose batteries fall under the Battery Waste Management Rules, 2022, which require consumers to discard them separately. What’s missing is a simple path for a household to follow.

SafeDrop targets the two blockers directly: **“I don’t know how to erase it”** becomes a device-specific checklist, and **“I don’t know who to give it to”** becomes a map of government-registered sites, a booking and a verifiable receipt.

WHAT SAFEDROP DOES

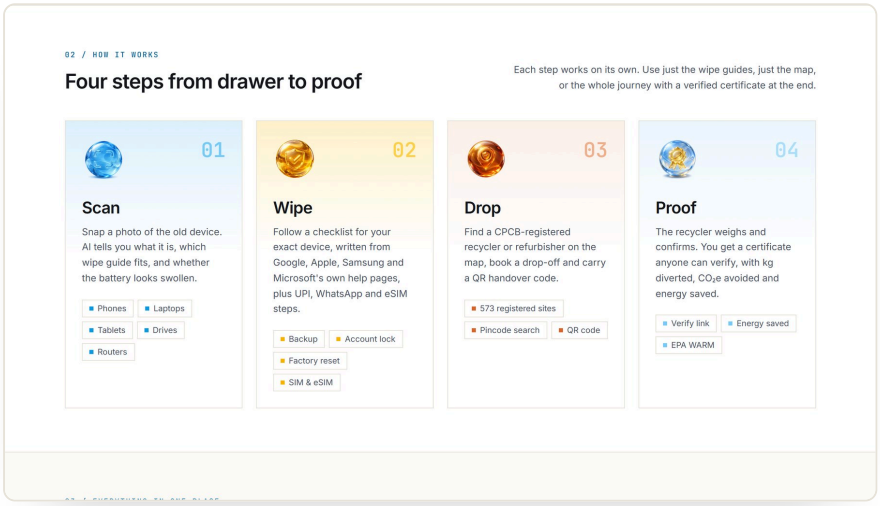
02 From drawer to proof in four steps

01
Scan
Photograph the device. AI names it, picks the wipe guide and flags a swollen battery.

02
Wipe
Tick through official steps. Critical ones (account locks, eSIM, SD card) must be done first.

03
Drop
Find a CPCB-registered site, book a drop-off or a kabadiwala doorstep pickup, and carry a QR code.

04
Proof
The drop point weighs and confirms. A public certificate shows kg, CO_2e and energy saved.



The home page's "How it works" section. Each step also works on its own: the wipe guides and the recycler map need no account.

Who it helps

WHO	WHAT THEY GET
Households	Data stays private, nothing to install, a clear place to hand devices over, and proof that it happened.
Kabadiwalas	Verified doorstep pickups in their own pincodes, and a registered drop point to deliver to instead of burning cables.
Registered recyclers	Wiped, sorted devices arriving with a record of what was handed over.
The city	Less open burning, metals recovered, and energy and CO_2e savings counted from real weighed handovers.

THE 15 FEATURES

03 Everything in one place

Every feature below was checked on the live AWS deployment, through the browser or the live API (chapter 13).

01 **AI device scan**
Photo in; device type, brand and matching wipe guide out.

02 **Swollen-battery check**
Flags risky lithium batteries and shows the safe-handling steps.

03 **10 official wipe guides**
104 steps for phones, laptops, drives, TVs and routers.

04 **India privacy steps**
UPI apps, WhatsApp backup, eSIM, Find My Device.

05 **Lock-out alerts**
Critical steps (Google FRP, Activation Lock) gate “Mark as wiped”.

06 **Ask SafeDrop**
AI answers in English or Hindi only from official steps, with sources.

07 **Registered recycler map**
413 active recyclers and 154 refurbishers on Amazon Location maps.

08 **Nearest first**
Sort by distance from GPS or a pincode, geocoded by Amazon Location.

09 **“Is it registered?”**
Check a collector’s company name against the CPCB list.

10 **Reuse first**
Refurbishers shown for working devices; reuse saves about 8× more energy.

11 **Drop-off or doorstep pickup**
Carry devices to a drop point or have a kabadiwala partner collect.

12 **QR handover code**
A 6-character code ties the wiped devices to the handover.

13 **Partner portals**
Drop points verify, weigh and confirm; kabadiwalas accept and log pickups.

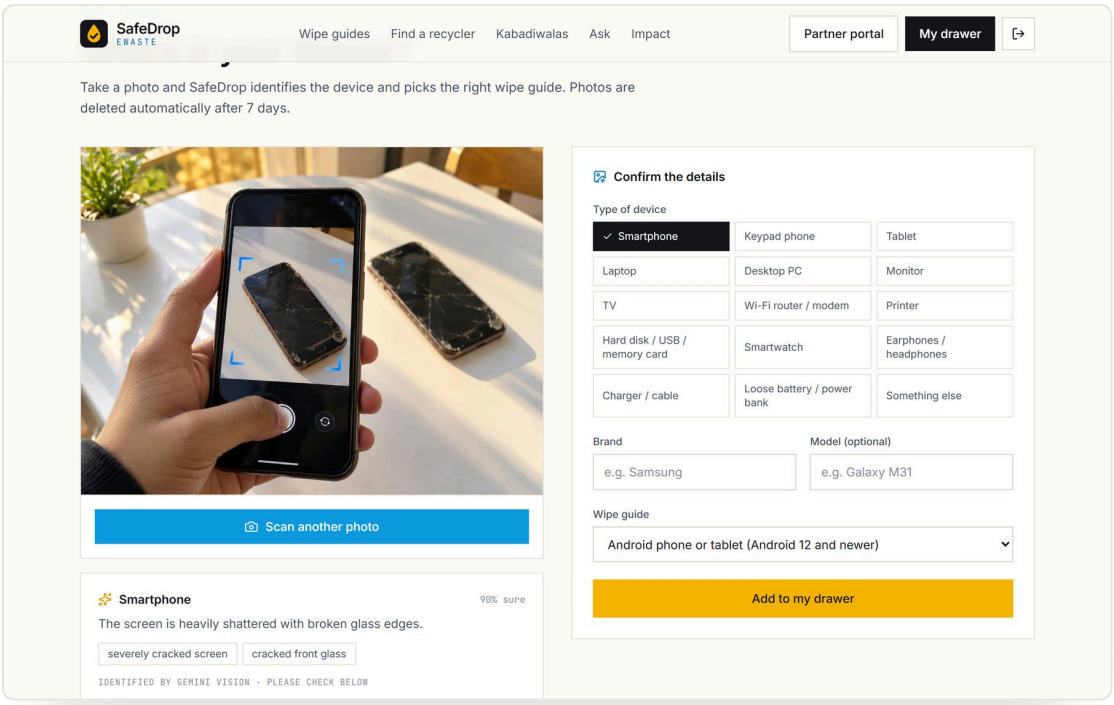
14 **Verifiable certificate**
A public link and QR prove the devices reached a drop point.

15 **Waste to energy saved**
Kg diverted, CO_2e avoided and kWh saved from US EPA WARM v16.

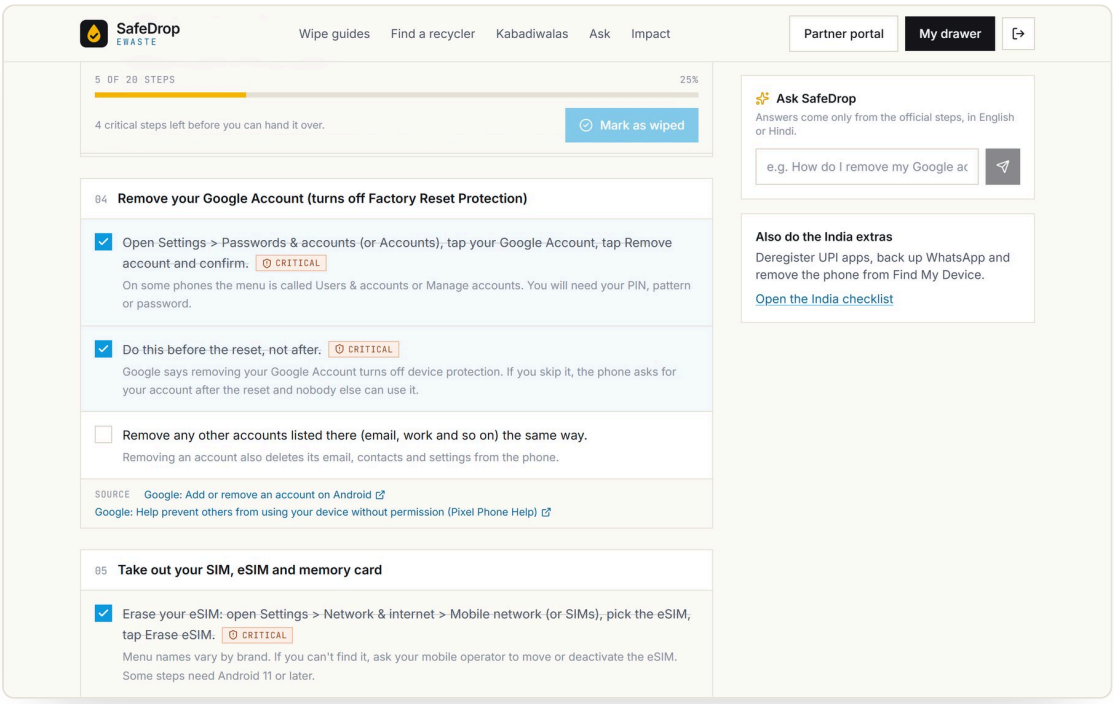
The track asks teams to *“close the loop on what a city throws away and cut what it burns”*. SafeDrop covers four of its listed problems directly (E-waste, Recycling, Segregation and Informal recyclers) and reports the energy each handover saves.

WALKTHROUGH

04 Scan and wipe



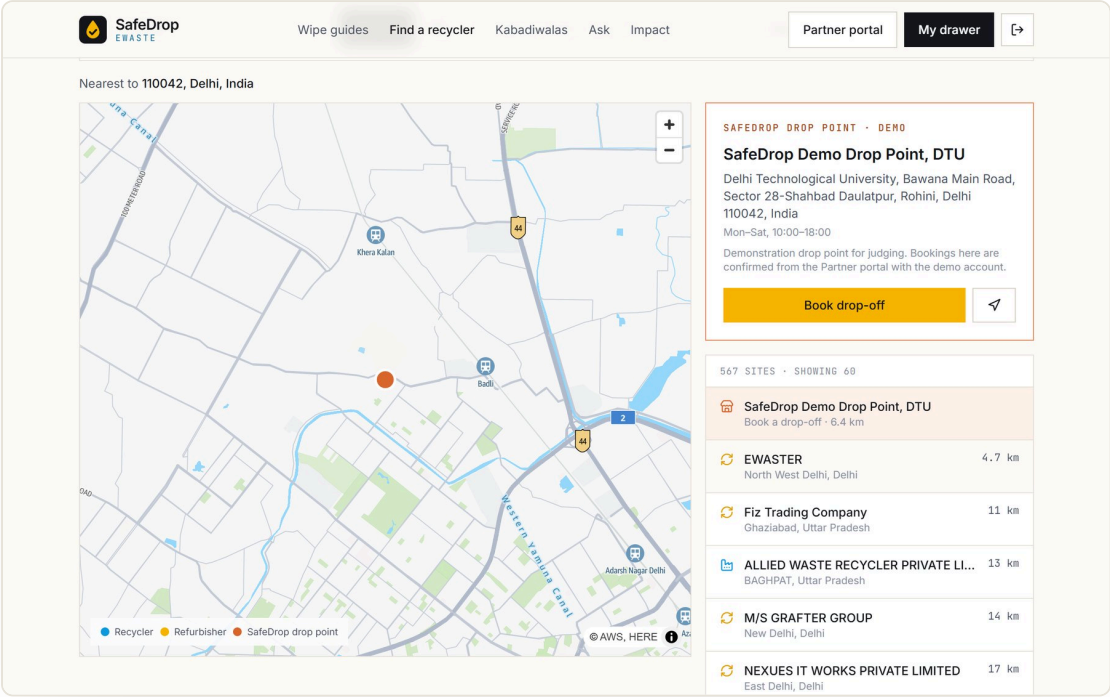
Add a device. The photo is resized in the browser, uploaded straight to S3 through a 5-minute presigned URL, and identified by Gemini vision. Amazon Rekognition takes over if Gemini is unavailable. The user confirms or corrects the type and brand, and the wipe guide follows the brand.



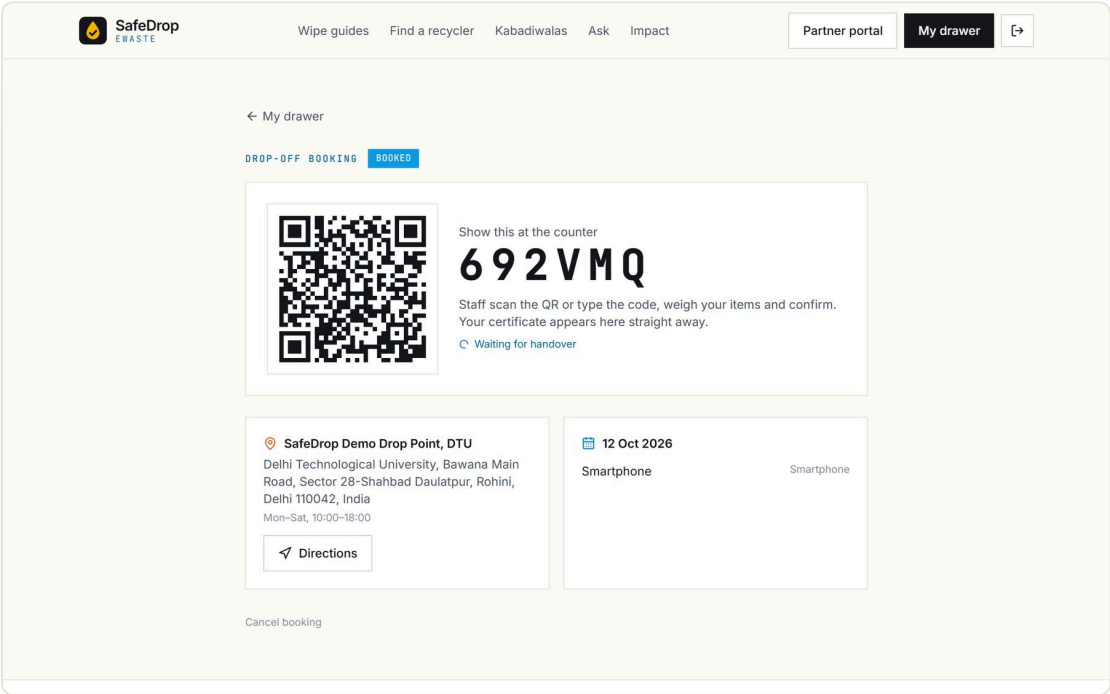
The wipe checklist. Progress is saved per step in DynamoDB. “Mark as wiped” stays disabled until every critical step is ticked, and the API enforces the same rule, so it can’t be skipped from outside the app.

WALKTHROUGH

05 Find a recycler and book



Nearest to pincode 110042. Amazon Location Service renders the map (India political view) and geocodes the pincode in the browser with a key that only works from the app's own domain. Gold pins are refurbishers, sky-blue pins are recyclers and the orange pin is the SafeDrop drop point.

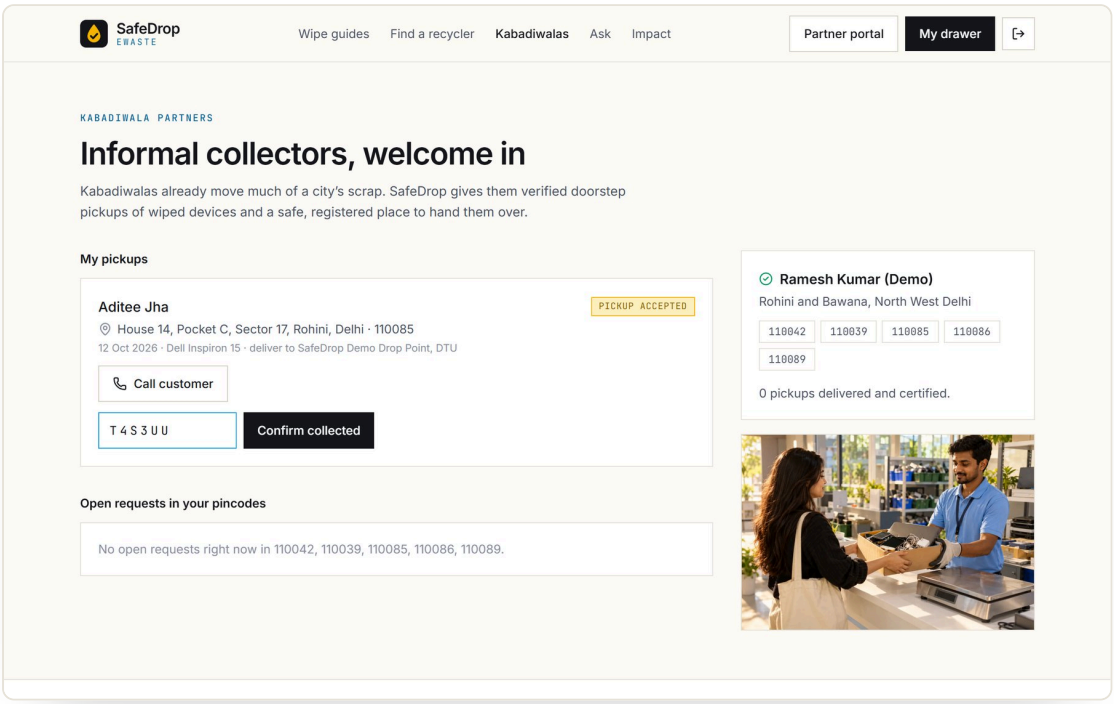


The booking. Creating it is one DynamoDB transaction: the booking, a unique handover code and every device's status change succeed or fail together. The page refreshes itself until the handover is confirmed.

INFORMAL RECYCLERS

06 Kabadiwala pickup partners

Kabadiwalas already move much of a city’s scrap. SafeDrop brings them into the formal chain instead of cutting them out.



The pickup partner dashboard. A collector applies with a phone number and the pincodes they work. After a verification call they are approved and see open requests in those pincodes. The customer’s address and phone are shared only after the collector accepts.

Chain of custody

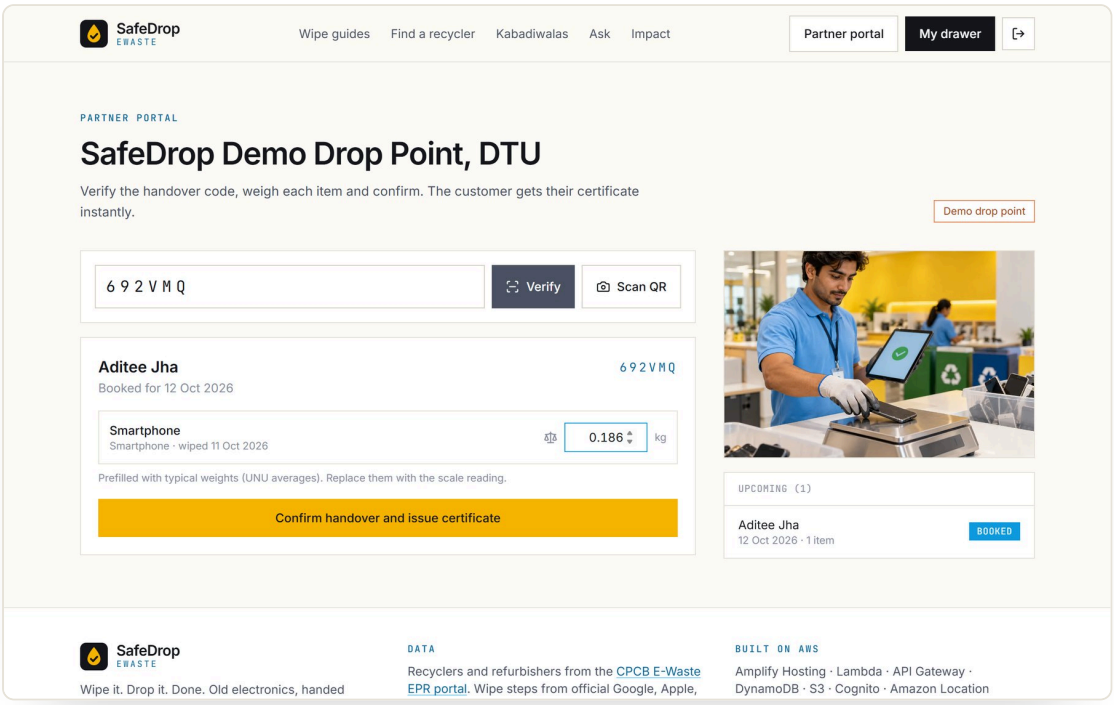
Request → accepted by the partner → collected (the partner types the customer’s code at the door) → received and weighed at the drop point. The drop point refuses the code until the pickup is marked collected.

What the customer sees

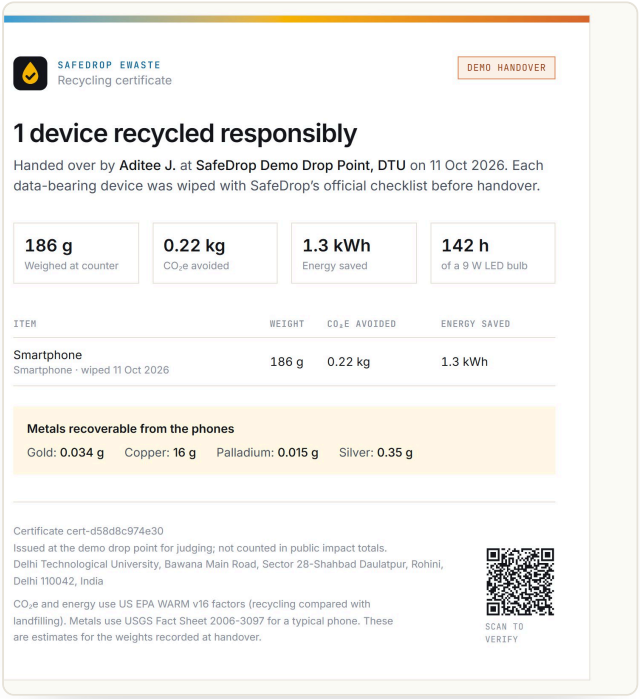
A live timeline with the partner’s name and a call button, and the same certificate as a drop-off, naming the pickup partner.

WALKTHROUGH

07 Handover and certificate



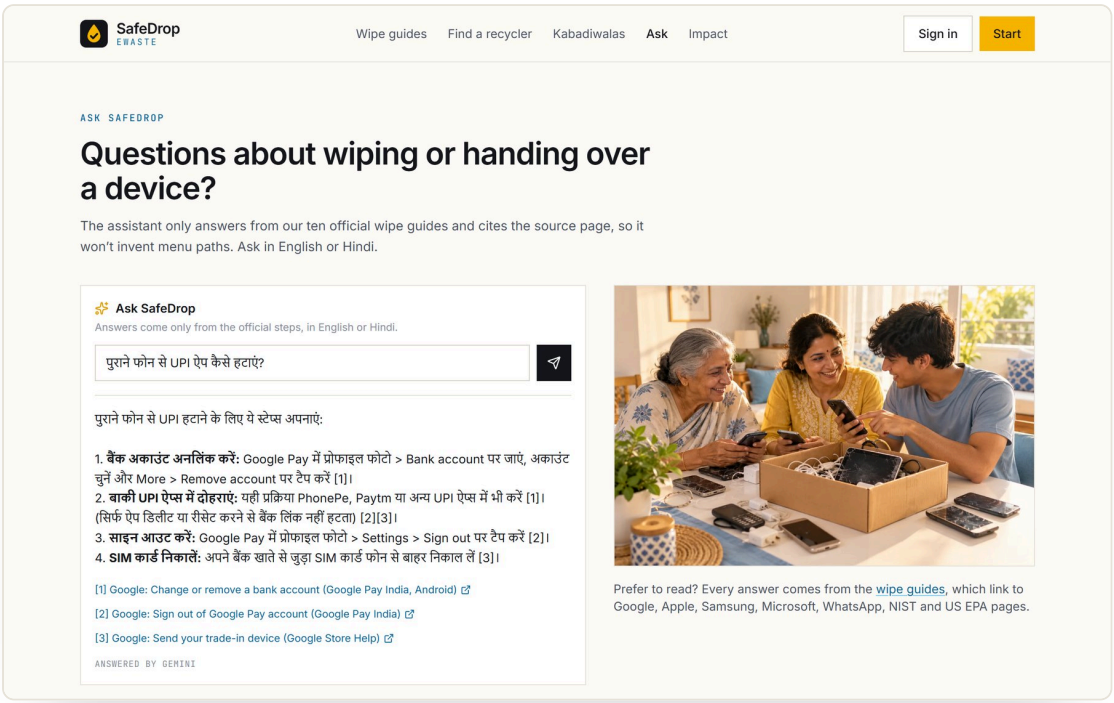
Partner portal. Staff scan the QR or type the code, see what was booked and whether each item was wiped, and enter the scale reading. Typical UNU weights prefill the form; the recorded weight is what counts.



The certificate. Anyone can open the public link or scan its QR. Only the holder's first name and last initial are shown. This one is labelled *Demo handover* because it was issued at the demo drop point.

AI

08 Ask SafeDrop: grounded AI



A Hindi question about UPI apps, answered in Hindi with numbered citations that match the Google help pages listed underneath.

How the answer is kept honest

- **Grounded:** the model receives only the relevant official guide steps and their numbered sources, and is told never to invent menu paths.
- **Cited:** citations are renumbered on the server so they match the source list shown to the user.
- **Resilient:** four Gemini Flash models are tried in turn, with a cooldown for any that is busy. If all fail, a keyword search over the guides answers instead. Amazon Bedrock (Nova Lite) can be switched on as the first choice once the account allows it.
- **Bounded:** 40 AI calls per user or IP per day, counted in DynamoDB with a 2-day TTL, so costs can't run away.

WASTE TO ENERGY

09 How impact is counted

Recovered copper, gold and aluminium replace metal that would otherwise be mined, smelted and refined. SafeDrop counts that saving from the weight recorded at each handover, using published US EPA factors.

```
CO₂e avoided (kg) = weight_kg ÷ 907.18 × (landfilling - recycling) [WARM v16, t CO₂e per short ton] × 1000
Energy saved (kWh) = weight_kg ÷ 907.18 × (landfilling - recycling) [WARM v16, million Btu per short ton] × 293.07
LED-bulb hours = kWh × 1000 ÷ 9 W
```

WARM material	SafeDrop device types	CO_2E, t per short ton	kWh per short ton
Portable Electronic Devices	Phones, keypad phones, tablets, laptops, smartwatches	1.08	6,219
Desktop CPUs	Desktop PCs	1.51	6,295
Flat-Panel Displays	Monitors, TVs	1.01	4,496
Hard-Copy Devices	Printers	0.58	2,397
Mixed Electronics	Routers, drives, earphones, chargers, other	0.92	4,188

Worked example: one phone

A 0.09 kg phone (UNU average) saves 0.09 ÷ 907.18 × 21.22 × 293.07 = **0.62 kWh**, about **69 hours** of a 9 W LED bulb, and avoids **0.11 kg CO_2e**. USGS puts the recoverable metal in a typical phone at 16 g copper, 0.35 g silver, 0.034 g gold and 0.015 g palladium.

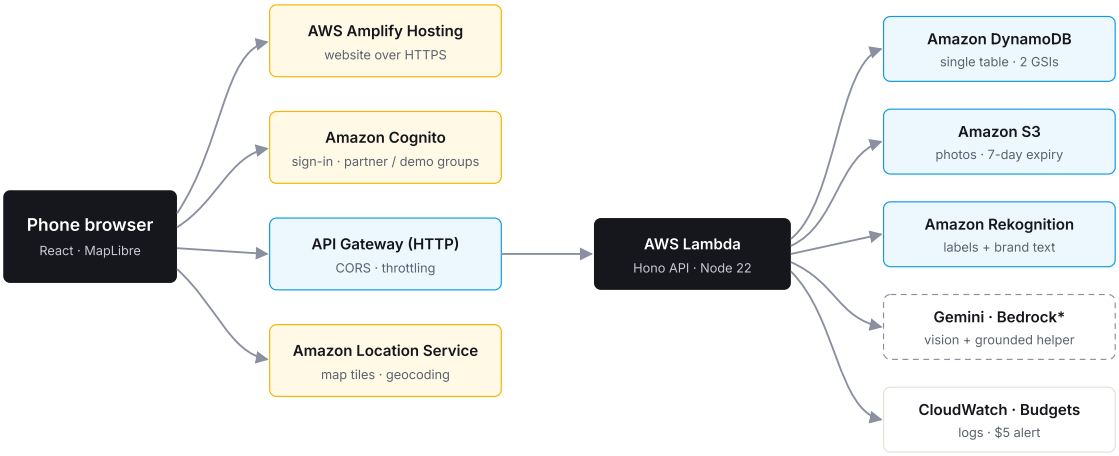
Reuse beats recycling

WARM credits *not making* a new portable device with 163.36 million Btu per short ton, about 8× the recycling credit of 20.95. That's why the map shows registered refurbishers for working devices.

Loose batteries are not covered by WARM's electronics factors, so they count towards weight only. Demo-account handovers go to a separate counter and never reach the public totals.

ARCHITECTURE

10 Built on AWS



Everything above is one AWS CDK stack (infra/lib/stack.ts), deployed to Mumbai (ap-south-1). * Bedrock is wired in and switched on with one flag once the account allows it.

AWS SERVICE	WHAT IT DOES IN SAFEDROP
AWS Lambda + API Gateway	The whole API: devices and wipe progress, bookings, pickups, partner handovers, certificates, AI endpoints. Preflights return 204 before auth; the stage is throttled.
Amazon DynamoDB	One table with two indexes (by user; by drop point, pickup pincode or collector). Bookings and handovers are atomic `TransactWrite`s.
Amazon Cognito	Email sign-up and sign-in. `partner` and `demo` groups and a `custom:pointId` attribute; the web client cannot write that attribute.
Amazon S3	Device photos via presigned PUT URLs, encrypted, public access blocked, deleted after 7 days by a lifecycle rule.
Amazon Location Service	Map tiles with the India political view; one-time geocoding of all 573 CPCB addresses; live pincode search from the browser.
Amazon Rekognition	Device type from labels, and brand from detected text, whenever the vision model is unavailable.
AWS Amplify Hosting	Serves the React site over HTTPS with single-page-app rewrites; each build is uploaded as a manual deployment.
AWS CDK, CloudWatch, Budgets	Infrastructure as TypeScript, one-week API logs, and a $5 monthly cost alert.

DESIGN CHOICES

11 Data model, privacy and safety

One DynamoDB table

ITEM	KEY	INDEX
Device	`USER#id` / `DEVICE#id`	—
Booking	`BOOKING#id` / `META`	gsi1: user · gsi2: drop point, pickup pincode or collector
Handover code	`CODE#XXXXXX` / `META`	Written with a uniqueness condition
Certificate	`CERT#id` / `META`	gsi1: user
Pickup partner, coverage	`COLLECTOR` / user · `COVER#pincode` / user	—
Counters, AI quota	`STATS` / `TOTAL` or `DEMO` · `QUOTA#id` / day	Quota rows expire by TTL

Privacy and safety by design

Personal data

Photos expire after 7 days. Public certificates show only a first name and last initial. A pickup partner sees the address and phone only after accepting. Handover codes skip look-alike characters (0/O, 1/l) and die after use.

Integrity

Critical wipe steps are enforced on the server. Bookings, pickups and handovers use conditional writes, so a device can't be booked twice and a code can't be confirmed twice.

Battery safety

Swollen batteries are flagged from the photo and on every phone, tablet and laptop guide, with US EPA and US Fire Administration handling steps.

Cost and abuse limits

API throttling, daily AI quotas, a referer-restricted map key and a budget alert keep a public demo from running up bills.

REAL DATA ONLY

12 Where every number comes from

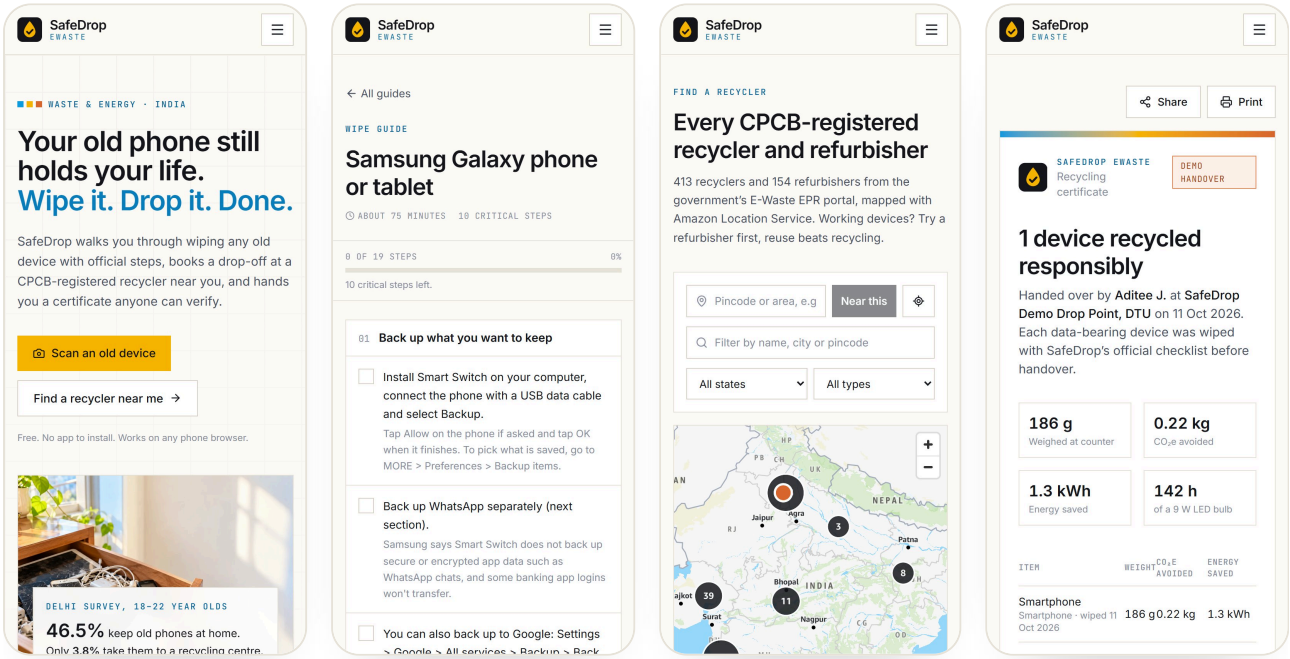
DATASET	SOURCE	SIZE	USED FOR
Registered recyclers 6 marked “operations disabled” are hidden	CPCB E-Waste EPR portal, registration-granted list, downloaded 10 Oct 2026	419	Map, nearest search, “Is it registered?”
Registered refurbishers	CPCB E-Waste EPR portal, same date	154	“Reuse first” on the map
Locations	Amazon Location Service geocoding, with each pin’s precision recorded	573	229 street address · 182 locality · 154 pincode · 8 district
Wipe guides	Google, Samsung, Apple, Microsoft, WhatsApp, HMD, TP-Link, HP and NIST SP 800-88 Rev. 2, read in October 2026	104 steps	Checklists and the AI helper’s only knowledge
Impact factors	US EPA WARM v16 (Dec 2023): Electronics Exhibit 1-5 and Energy Impacts Exhibit 7-12	7 materials	CO_2e and kWh on certificates
Typical weights	UNU E-waste Statistics guidelines (Forti et al., 2018), via UNEP	11 types	Prefilling the partner’s scale form only
Context facts	PIB/CPCB, Global E-waste Monitor 2024, WHO, peer-reviewed surveys, E-Waste and Battery Rules 2022	45	Home and impact pages

The portal list has one row per registration, so some companies appear twice. CPCB’s own press note counted 322 registered recyclers as on 9 Feb 2025; the October 2026 list has 419 registration rows. Every pin shows how precisely it was located, and the app tells users to call ahead.

TESTING

13 Testing and validation

TEST	WHAT IT COVERS	RESULT
Unit tests `npm test`	Handover codes, guide matching, CO_2e and energy maths against EPA factors, metals, totals, certificate name shortening, citation renumbering	9 / 9 PASS
Live end-to-end `scripts/e2e.mjs`	Against the deployed API: sign-in, wipe enforcement, drop-off booking, partner handover, public certificate, code reuse blocked, full kabadiwala pickup chain, AI helper, photo identification	10 / 10 PASS
Browser flows	Microsoft Edge driving the live site at 1366 px and 390 px: sign in, add device, tick critical steps, mark wiped, book, partner confirm, open certificate	PASS, NO PAGE ERRORS
Feature checks	Swollen-battery photo flagged; “Attero” found and a fake name warned; GPS in Delhi lists the nearest recycler at 6.8 km	PASS



On a 390 px phone screen: home, Samsung wipe guide, recycler map and certificate. Every page was checked for sideways overflow.

HONEST LIMITS

14 Limits and next steps

TODAY	NEXT
One demo drop point (at DTU, Delhi) runs the partner flow. Other registered recyclers are listed with their published phone and email.	Let CPCB-registered recyclers claim their listing with their registration details and run their own counter.
Pickup partners are approved by hand after a call (`approve-collector.mjs`).	An admin screen, partner ratings and an agreed per-kg collection fee.
The new AWS account is still being verified, so CloudFront and Bedrock are blocked. The site runs on Amplify Hosting and the AI on Gemini.	Switch Bedrock Nova on with `BEDROCK_ENABLED=true` once verification completes.
The interface is in English; the AI helper answers Hindi questions in Hindi.	Hindi and other Indian languages for the whole interface and the guides.
Impact uses US EPA WARM factors, the best published per-device factors found.	Use Indian recycler process data once partners share it.

What already works today: a household can wipe a device with official steps, find the nearest registered site, hand it over at a partner point or through a verified kabadiwala, and get a certificate anyone can check, all on AWS and all from real data.

REFERENCES

15 Sources

Screenshots in this document are from the live deployment on 11 October 2026. Product images on the website are AI-generated illustrations, labelled as such where they depict real-world scenes.

1. Press Information Bureau (MoEFCC), CPCB e-waste generation FY 2023-24 and 2024-25, 11 Aug 2025. https://pib.gov.in/PressReleasePage.aspx?PRID=2155124
2. Press Information Bureau (MoEFCC), e-waste collected and processed, 24 Jul 2025. https://pib.gov.in/PressReleasePage.aspx?PRID=2147876
3. Press Information Bureau (MoEFCC), registered recyclers and refurbishers as on 09.02.2025, 13 Feb 2025. https://pib.gov.in/PressReleasePage.aspx?PRID=2102701
4. CPCB E-Waste EPR portal, registration-granted lists of recyclers and refurbishers. https://eprewaste.cpcb.gov.in/
5. Borthakur A., Singh P. (2022), survey of 1,039 young adults in New Delhi, Energy, Ecology and Environment. https://pmc.ncbi.nlm.nih.gov/articles/PMC8853223/
6. YouGov (May 2024), What do Britons do with their old phones? https://yougov.com/en-gb/articles/50080-what-do-britons-do-with-their-old-phones-half-of-t
7. World Health Organization, Electronic waste (e-waste) fact sheet, 1 Oct 2024. https://www.who.int/news-room/fact-sheets/detail/electronic-waste-(e-waste)
8. Rajya Sabha Secretariat (2011), E-waste in India. https://cms.rajyasabha.nic.in/UploadedFiles/ElectronicPublications/E-Waste_in_india.pdf
9. MoEFCC, E-Waste (Management) Rules, 2022, G.S.R. 801(E). https://cpcb.gov.in/uploads/Projects/E-Waste/e-waste_rules_2022.pdf
10. MoEFCC, Battery Waste Management Rules, 2022, S.O. 3984(E). https://egazette.gov.in/WriteReadData/2022/238351.pdf
11. ITU and UNITAR, Global E-waste Monitor 2024. https://ewastemonitor.info/the-global-e-waste-monitor-2024/
12. US EPA, WARM v16 Electronics chapter (EPA-530-R-23-022), Dec 2023. https://www.epa.gov/system/files/documents/2023-12/warm_electronics_v16_dec.pdf
13. US EPA, WARM v16 Management Practices chapters, Chapter 7 Energy Impacts, Dec 2023. https://www.epa.gov/system/files/documents/2024-01/warm_management_practices_v16_dec.pdf
14. NIST SP 811, Guide for the Use of the International System of Units, Appendix B.8. https://www.nist.gov/pml/special-publication-811/nist-guide-si-appendix-b-conversion-factors/nist-guide-si-appendix-b8
15. US Geological Survey, Fact Sheet 2006-3097, Recycled Cell Phones. https://pubs.usgs.gov/fs/2006/3097/fs2006-3097.pdf
16. US EPA, Electronics Donation and Recycling. https://www.epa.gov/recycle/electronics-donation-and-recycling
17. US EPA, Used Lithium-Ion Batteries. https://www.epa.gov/recycle/used-lithium-ion-batteries
18. Forti V., Baldé C. P., Kuehr R. (2018), E-waste Statistics: Guidelines on Classification, Reporting and Indicators, 2nd ed., UNU; average weights via UNEP. https://sdgs.unep.org/article/annex-v-indication-average-weight-electric-and-electronic-equipment-eu-28-kgpiece
19. NIST SP 800-88 Rev. 2, Guidelines for Media Sanitization (Sept 2025), and vendor help pages listed in the app next to each wipe step.